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Electrical Characterization of ALD1106 and organic FETs using current-driven gated van der Pauw's method

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Abstract

1. **1.** Complete electrical characterization of commercial n-MOSFET (ALD1106): transfer and output characteristics (good agreement with the datasheet).
   Extraction of charge-carrier mobility ($\mu = (760 \pm 2)\ \mathrm{cm^2V^{-1}s^{-1}}$) and background resistance ($R_\infty = (65 \pm 2)\ \Omega$) using C-gvdP method.
2. **2.** Complete electrical characterization of organic FETs (Sample: PS9 and PS12): transfer and output characteristics.
   The intrinsic charge-carrier mobility and background resistance were determined using the C-gvdP method:

   PS9: $\mu = (3.55 \pm 0.04)\ \mathrm{cm^2V^{-1}s^{-1}}$, $R_\infty = (4.3 \pm 0.1)\ \mathrm{M\Omega}$.
   PS12: $\mu = (3.44 \pm 0.05)\ \mathrm{cm^2V^{-1}s^{-1}}$, $R_\infty = (36.4 \pm 0.1)\ \mathrm{M\Omega}$.
3. **3.** Discussion of a possible origin of the large background resistance R_∞.

Theory

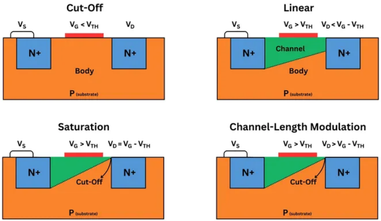

Figure: Operation modes of n-MOSFET. (Source: Internet)

$$I_{\mathrm{DS}} = \begin{cases} 0, & \text{Cut-off, } V_{\mathrm{GS}} < V_{\mathrm{th}} \\ \mu C_i \dfrac{W}{L}\left[(V_{\mathrm{GS}} - V_{\mathrm{th}}) V_{\mathrm{DS}} - \dfrac{V_{\mathrm{DS}}^2}{2} \right], & \text{Linear, } V_{\mathrm{GS}} > V_{\mathrm{th}},\ V_{\mathrm{DS}} < V_{\mathrm{GS}} - V_{\mathrm{th}} \\ \dfrac{1}{2}\mu C_i \dfrac{W}{L} (V_{\mathrm{GS}} - V_{\mathrm{th}})^2, & \text{Saturated, } V_{\mathrm{GS}} > V_{\mathrm{th}},\ V_{\mathrm{DS}} \geq V_{\mathrm{GS}} - V_{\mathrm{th}} \end{cases}$$

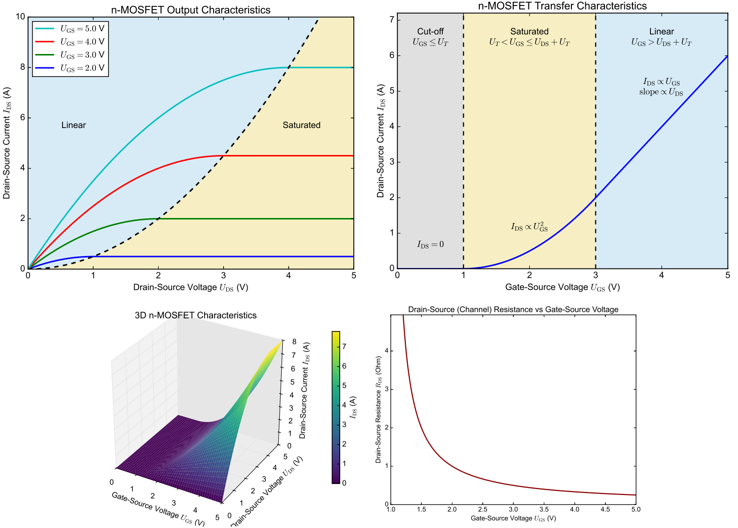

Figure: Theoretical I-V Characteristics of n-MOSFET

I-V Characteristics of ALD1106

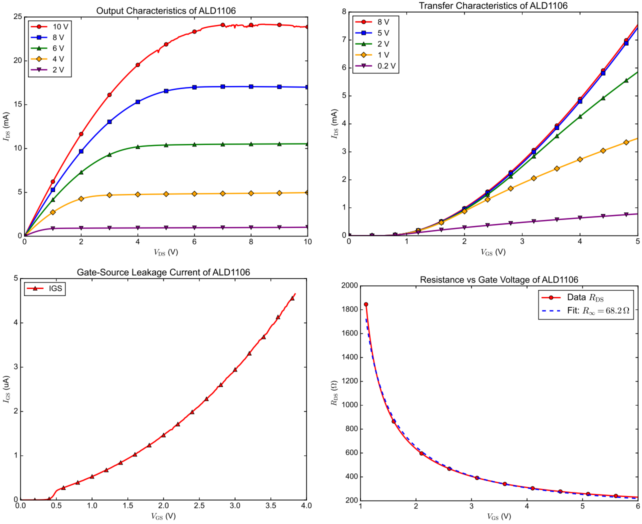

Figure: I-V Characteristics of ALD1106

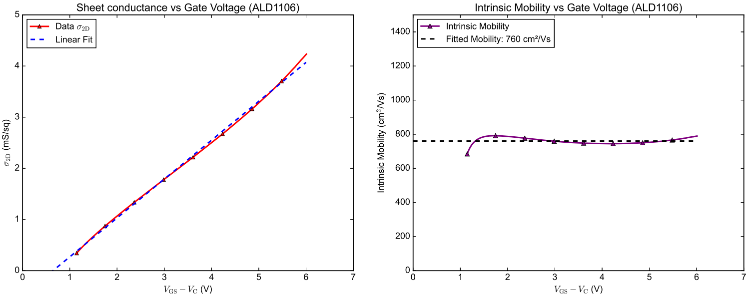

Figure: Extraction of threshold voltage, charge-carrier mobility of ALD1106

Fit results: $\sigma = \mu C_i(V_{GS} - V_C - V_T) \longrightarrow y = ax + b$
(whereas $y = \sigma$, $x = V_{GS} - V_C$, $V_C = (V_1 + V_2)/2$, $C_i = 10^{-2}\ \mathrm{F\,m^{-2}}$).

$$\mu = \frac{a}{C_i} \cdot 10^4 = (760 \pm 2)\ \mathrm{cm^2V^{-1}s^{-1}},$$

$$V_T = -\frac{b}{a} = (0.644 \pm 0.009)\ \mathrm{V},$$

$$r^2 = 0.998.$$

C-gvdP for ALD1106

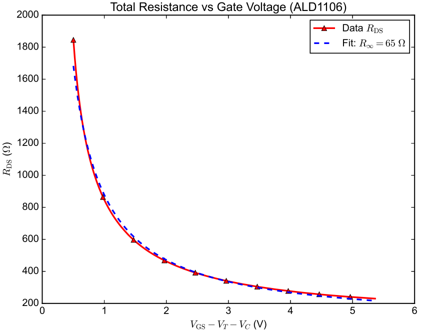

Figure: Extraction of background resistance of ALD1106

Fit results: $R_{\mathrm{DS}} = \dfrac{V_{\mathrm{DS}}}{I_{\mathrm{DS}}} = R_\infty + \dfrac{b}{V_{\mathrm{GS}} - V_{\mathrm{C}} - V_{\mathrm{T}}} \longrightarrow y = a + \dfrac{b}{x}.$

$$R_\infty = (65 \pm 2)\ \Omega,$$
$$r^2 = 0.996.$$

I-V Characteristics of PS9 and PS12

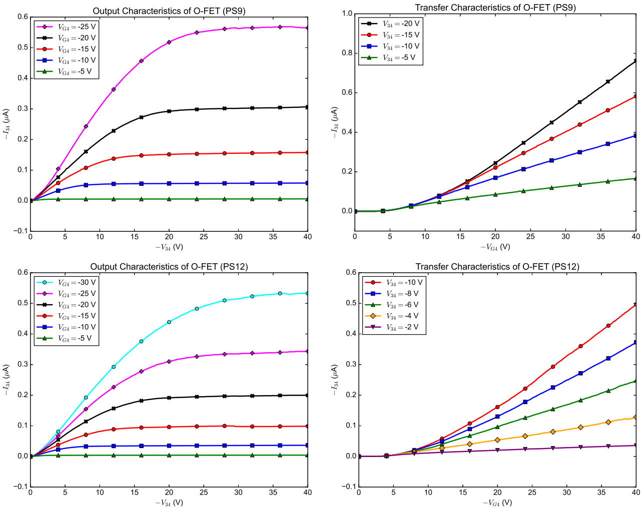

Figure: I-V Characteristics of Sample PS9 and PS12

C-gvdP for PS9

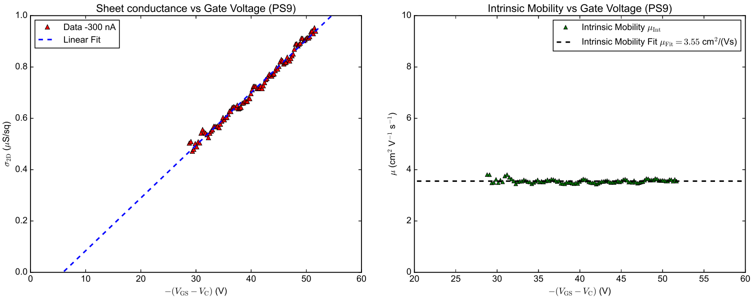

Figure: Extraction of V_T, μ of PS9

Fit results:

$$\mu = (3.55 \pm 0.04)\ \mathrm{cm^2V^{-1}s^{-1}},$$
$$V_T = (-5.9 \pm 0.4)\ \mathrm{V},$$
$$r^2 = 0.991.$$

C-gvdP for PS9

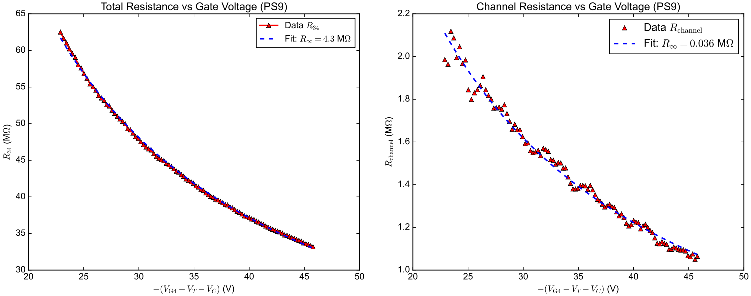

Figure: Extraction of background of total and channel resistance of PS9

Fit results:

$$R_\infty = (4.3 \pm 0.1)\ \mathrm{M\Omega},$$
$$r^2 = 0.999.$$

C-gvdP for PS12

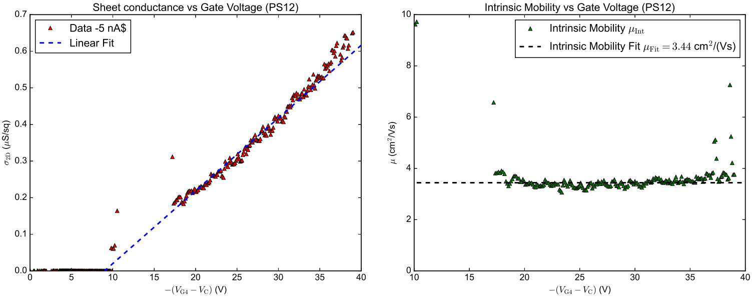

Figure: Extraction of V_T, μ of PS12

Fit results:

$$\mu = (3.44 \pm 0.05)\ \mathrm{cm^2V^{-1}s^{-1}},$$
$$V_T = (-9.0 \pm 0.4)\ \mathrm{V},$$
$$r^2 = 0.876.$$

C-gvdP for PS12

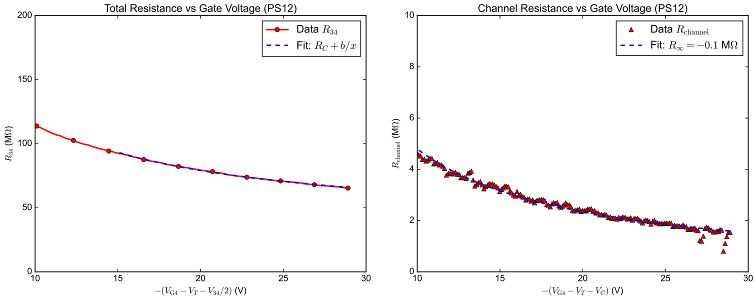

Figure: Extraction of background of total and channel resistance of PS12

Fit results:

$$R_\infty = (36.4 \pm 0.1)\ \mathrm{M\Omega},$$
$$r^2 = 0.999.$$

Possible origin of large R_∞

The background resistance of PS9 and PS12 is on the order of MΩ, much larger than for ALD1106.

In organic FETs, the measured two-terminal resistance can be dominated by the metal/organic contact.

A mismatch between the electrode work function and the HOMO level can create a Schottky-like injection barrier for holes.

This increases contact resistance and leads to poor charge injection.

Additional contributions may come from interface traps, structural disorder, or imperfect contact formation.

Advantage of C-gvdP

The total device resistance contains both channel transport and parasitic contact effects.

The current-driven gated van der Pauw method separates the channel conductance from these contact-related resistances.

Thank you for your attention!
Any questions?